Cambridge Trust Scholarship Essay

Candidate: Suleyman Hajizadeh | Western Caspian University, Azerbaijan

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Azerbaijan has no bioinformatics training programme. The country has medical schools, biology departments, and a growing interest in data science — but the computational infrastructure to connect molecular biology with quantitative analysis does not yet exist at scale. A student who wants to learn RNA-seq analysis learns it alone: from papers, from online courses, from trial and error on a personal laptop.

This is not a complaint. It is a problem worth solving — and it is the problem that motivates my Cambridge application.

I am that student. My work on triple-negative breast cancer (TNBC) transcriptomics — an end-to-end Snakemake pipeline from raw SRA reads through DESeq2 differential expression and nested cross-validated subtype classification — was built without a supervisor, a high-performance compute cluster, or a local community of practice. My implementation of the Kimura 2-Parameter (K2P) phylogenetic distance model and Neighbour-Joining tree algorithm was written from the mathematical equations in the original Kimura (1980) paper, not from a library call. My Miyazawa-Jernigan contact potential analysis of AKT1 structural stability was conducted using PDB coordinate files parsed by hand.

I have learned, by necessity, to be a self-contained research unit. But self-containment is a constraint, not a virtue. The questions I want to ask — why does TNBC tumour heterogeneity confound bulk RNA-seq subtype signatures? What Bayesian model best captures the overdispersion structure of low-replicate cancer transcriptomes? — require a research environment where I can push beyond the methodological frontier, not reconstruct it.

Cambridge offers that environment. Dr. Sarah Teichmann's group at the Wellcome Sanger Institute (now relocated to Cambridge) pioneered single-cell atlas construction precisely in the tumour types I study. The MRC Biostatistics Unit's work on Bayesian hierarchical models for genomic data is the statistical framework I need to extend my Negative Binomial modelling. These are not generic research environments I am naming for effect — they are the specific groups whose published methods I have been reading, implementing, and building upon for two years.

A Cambridge PhD is not the end of my connection to Azerbaijan; it is how I acquire the expertise to return and build what does not yet exist: a rigorous, reproducible, mathematically grounded approach to computational biology research in my home country.

The infrastructure gap is real. I intend to close it.

Word count: ~390